

JUNOH JUNG

Postdoctoral Appointee | Mathematics and Computer Science Division | Argonne National Laboratory

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RESEARCH PROFILE

Computational scientist developing **physics-machine-learning methods** for accurate, scalable modeling, prediction, and control of fluid flows. Research connects **differentiable simulation and high-performance computing (HPC)** with flow estimation and control to improve predictive accuracy and computational efficiency.

RESEARCH FUNDING AND COMPUTING ALLOCATIONS

Argonne Laboratory Directed Research and Development (LDRD) Sole Principal Investigator Research award: \$25,000 . Project: Agent-Orchestrated Multi-Fidelity Workflow for Stable Hybrid Physics-Machine Learning Simulations.	Apr-Sep 2026
ALCF Director's Discretionary Allocation Program Argonne Leadership Computing Facility (ALCF). Sole Principal Investigator Computing allocation: 90,000 node-hours over 12 months . Project: Differentiable Hybrid Physics Machine Learning Simulations.	2026

EDUCATION

Ph.D., Mechanical Engineering, University of Michigan Dissertation: Resolvent-based Estimation and Control of Aerodynamic Flows. Advisor: Aaron Towne.	2019-2024
M.S., Aerospace Engineering, RWTH Aachen University Thesis: Development of a fully coupled zonal RANS/LES method for the simulation of a turbulent backward-facing step flow. Advisor: Wolfgang Schroeder.	2014-2018
B.S., Mechanical and System Design Engineering, Hongik University Graduating rank: 1 of 166 in the department.	2008-2014

RESEARCH APPOINTMENTS AND EXPERIENCE

Postdoctoral Appointee, Argonne National Laboratory Mathematics and Computer Science Division. Supervisor: Emil Constantinescu; Mentor: Bethany Lusch. Developing differentiable and hybrid physics-machine-learning methods for scalable flow simulation, including Diff-NekRS, weak-form corrections for finite-element models, and adaptive multi-fidelity workflows.	Nov 2024-present
Graduate Student Research Assistant, University of Michigan Mechanical Engineering. Advisor: Aaron Towne. Developed resolvent-based methods for estimation and control of aerodynamic flows, including laminar airfoil wakes and turbulent wakes.	Aug 2019-Aug 2024
Givens Associate, Argonne National Laboratory Mathematics and Computer Science Division. Supervisor: David Lenz; PI: Thomas Peterka. Researched physics-informed B-spline methods for flow-data reconstruction.	2023
Master's Thesis Researcher, RWTH Aachen University Aerodynamisches Institut Aachen (AIA). Advisor: Wolfgang Schroeder. Developed a fully coupled zonal RANS/LES method for turbulent backward-facing-step flow.	2017-2018
Visiting Scholar, Georgia Institute of Technology Department of Aerospace Engineering. Supervisor: Dimitri Mavris.	2017

PUBLICATIONS AND MANUSCRIPTS

Peer-reviewed Journal Articles

1. **Jung, J.**, Lenz, D., Constantinescu, E., Peterka, T. (2026). Physics-Informed B-spline Reconstruction of Flow Data. *Computer Methods in Applied Mechanics and Engineering*. doi: 10.1016/j.cma.2026.119436.
2. **Jung, J.**, Towne, A. (2026). Resolvent-based estimation of a turbulent wake. *Journal of Fluid Mechanics*, 1033, A22. doi: 10.1017/jfm.2026.11444.
3. **Jung, J.**, Bhagwat, R., Towne, A. (2025). Resolvent-based estimation and control of a laminar airfoil wake. *Journal of Fluid Mechanics*, 1016, A41. doi: 10.1017/jfm.2025.10423 (JFM Emerging Scholar Best Paper Prize nomination).
4. Farghadan, A., **Jung, J.**, Bhagwat, R., Towne, A. (2024). Efficient harmonic resolvent analysis via time stepping. *Theoretical and Computational Fluid Dynamics*, 38, 331–353. doi: 10.1007/s00162-024-00694-1.
5. Martini, E., **Jung, J.**, Cavalieri, A. V. G., Jordan, P., Towne, A. (2022). Resolvent-based tools for optimal estimation and control via the Wiener–Hopf formalism. *Journal of Fluid Mechanics*, 937, A19. doi: 10.1017/jfm.2022.102.

Preprints

1. **Jung, J.**, Balin, R., Lusch, B., Constantinescu, E. (2026). Diff-NekRS: A Scalable Differentiable Framework for Multi-Timestep Solver-in-the-Loop Training. arXiv preprint.

Conference Papers, Proceedings, and Abstracts

1. **Jung, J.**, Constantinescu, E., Balin, R., Lusch, B. (2026). A hybrid physics–machine-learning framework for enhancing a coarse-grid spectral element solver for large-scale computing. AIAA Aviation Forum. (Best Paper Award nomination).
2. **Jung, J.**, Constantinescu, E. (2026). Learning Differentiable Weak-Form Corrections to Accelerate Finite Element Simulations. ASME FEDSM.
3. Zhou, Y., Towne, A., **Jung, J.**, Bhagwat, R., Martini, E., Jordan, P., Audiffred, D., Maia, I., Cavalieri, A. V. G. (2026). Towards resolvent-based estimation and control of wavepackets in supersonic turbulent jets. AIAA SciTech Forum.
4. Towne, A., Bhagwat, R., Zhou, Y., **Jung, J.**, Martini, E., Jordan, P., Audiffred, D. B. S., Maia, I., Cavalieri, A. V. G. (2024). Resolvent-based estimation of wavepackets in turbulent jets. AIAA/CEAS 2024.
5. Towne, A., **Jung, J.**, Bhagwat, R., Zhou, Y., Paulson, J. (2024). Resolvent-based estimation and control of wakes and jets. 13th International Symposium on Turbulence and Shear Flow Phenomena.
6. **Jung, J.**, Towne, A. (2024). Toward turbulent wake estimation using a resolvent-based approach. AIAA SciTech Forum.
7. **Jung, J.**, Bhagwat, R., Towne, A. (2023). Resolvent-based estimation of laminar flow around an airfoil. AIAA SciTech Forum.

Forthcoming Conference Contribution

1. **Jung, J.**, Constantinescu, E., Balin, R. (2027). Agent-Orchestrated Multi-Fidelity Workflow for Stable Hybrid Physics–Machine Learning Simulations. AIAA SciTech Forum (*forthcoming*).

Manuscripts in Preparation

1. **Jung, J.**, Balin, R., Constantinescu, E. Agent-orchestrated multi-fidelity workflow for a stable hybrid physics-machine learning simulation.
2. **Jung, J.**, Balin, R., Lusch, B., Constantinescu, E. Diff-NekRS: Differentiable Hybrid Modeling for Spectral-Element Simulations.
3. **Jung, J.**, Constantinescu, E. Structure-Preserving Neural Variational Correction Operators for Hybrid Finite-Element Machine-Learning Simulations.

HONORS, FELLOWSHIPS, AND RECOGNITION

ALCF Postdoctoral Fellowship, Argonne National Laboratory	Nov 2024
Selected Participant, Argonne Training Program on Extreme-Scale Computing ATPESC; funded by the U.S. Department of Energy.	Aug 2023
Rackham Travel Grant, University of Michigan	Winter 2022; Winter 2023
Mechanical Engineering Fellowship, University of Michigan	Winter 2020
Merit-based Academic Scholarships, Hongik University Academic excellence scholarships, including full-tuition awards.	2009–2014

LEADERSHIP, MENTORING, AND TEACHING

Mentor and Participant, Argonne GPU Hackathons	2025–2026
Informal Mentor, Yuhao Zhou (Ph.D. student)	2022–2025
Graduate Student Instructor, University of Michigan	2024
Introduction to Fluid Mechanics (ME 320). Supported 101 students across two course sections (ME 320-001 and ME 320-002) through office hours, recitations, quiz review, and exam grading.	
Elected Local President	Oct 2014–Oct 2016
Korean-German Scientists and Engineers Association. Elected to represent the Aachen, Cologne, and Bonn region.	

PROFESSIONAL SERVICE

Session Chair, APS Division of Fluid Dynamics Meeting	2026 (upcoming)
Proposal Reviewer, APEX Program, Argonne National Laboratory	2026
Journal Referee	2026
Journal of Fluid Mechanics; Journal of Computational Physics; Computer Methods in Applied Mechanics and Engineering; Scientific Reports.	
Abstract Reviewer, AIAA SciTech and Aviation	2024–2026
Reviewed fluid dynamics submissions.	

Professional Memberships

Association for Computing Machinery (ACM), 2026; American Institute of Chemical Engineers (AIChE), 2026; American Society of Mechanical Engineers (ASME), 2026.
American Institute of Aeronautics and Astronautics (AIAA), 2022–2026; American Physical Society (APS), 2020–2026.

COLLABORATIVE PROPOSAL DEVELOPMENT

Key Personnel, Genesis Mission Phase 1 Proposal	Apr 2026
Led by Arizona State University; not selected. Project: AI-Enabled Design of Electrolyte Wetting Processes in Thick Porous Battery Electrodes. Focus: AI-enabled scientific workflows and adaptive learning methods for energy applications.	

TECHNICAL EXPERTISE

Computational methods: Computational fluid dynamics (DNS, LES, RANS); spectral-element and finite-element methods; flow estimation and control; DMD, POD, SPOD, and resolvent analysis.
Scientific machine learning: Differentiable simulation (Diff-NekRS); physics-informed B-spline reconstruction; weak-form corrections; deep learning (CNNs, GNNs, U-Net, Transformers); agentic simulation workflows.
Programming and HPC: C/C++, Python, Fortran; MPI (OpenMPI, MPICH); GPU and heterogeneous computing; PETSc, FFTW, LAPACK.
Tools and computing environments: Git/GitHub, Ubuntu, Bash; DOE, DoD, and university supercomputing environments; Tecplot, ParaView, VisIt; UG-NX and AutoCAD.

ADDITIONAL PROFESSIONAL ACTIVITIES

Invited Participant, Hyundai Motors Vision Conference	Aug 2022
Invited to South Korea for discussions with UAM engineers and factory visits.	
Sergeant, Republic of Korea Army	Feb 2010–Dec 2011

INVITED TALKS

1. **Jung, J.** (2027). From weak-form correction to Agent-orchestrated workflow for hybrid physics-ML framework, SIAM CSE27. Minisymposium: "Recent Advances in Galerkin Methods for Physical Systems" (*forthcoming*).
2. **Jung, J.** (2024). Estimation and Control of Aerodynamic Flows Using a Resolvent-Based Approach, ALCF.

SELECTED TALKS AND PRESENTATIONS

1. **Jung, J.**, Constantinescu, E., Balin, R. (2027). Agent-Orchestrated Multi-Fidelity Workflow for Stable Hybrid Physics–Machine Learning Simulations. AIAA SciTech Forum (*forthcoming*).
2. **Jung, J.**, Constantinescu, E. (2026). Towards agentic workflows for automated and stable hybrid physics–ML simulations. APS Division of Fluid Dynamics, oral presentation (*forthcoming*).
3. **Jung, J.**, Constantinescu, E. (2026). A hybrid physics–machine-learning framework for enhancing a coarse-grid spectral element solver for large-scale computing. Nek User Meeting.
4. **Jung, J.**, Constantinescu, E., Balin, R., Lusch, B. (2026). A hybrid physics–machine-learning framework for enhancing a coarse-grid spectral element solver for large-scale computing. AIAA Aviation.
5. **Jung, J.**, Constantinescu, E. (2026). Learning Differentiable Weak-Form Corrections to Accelerate Finite Element Simulations. ASME FEDSM, oral presentation.
6. **Jung, J.**, Constantinescu, E. (2026). Differentiable Weak-Form Learning for Coarse-Mesh Incompressible Flow Simulation: Toward Chemical Engineering Applications, 18th AIChE Midwest Regional Conference, oral presentation.
7. **Jung, J.**, Constantinescu, E. (2026). Differentiable Weak Form Learning for Structure Preserving Finite Element Acceleration of Incompressible Flows. USNCTAM26, oral presentation.
8. **Jung, J.**, Constantinescu, E. (2025). Hybrid Physics-Machine Learning Framework Toward Efficient Simulation of Turbulent Flows on an Exascale Platform. APS Division of Fluid Dynamics, oral presentation.
9. **Jung, J.**, Constantinescu, E. (2025). Hybrid Physics-Machine Learning Modeling with post-update discrete corrections. Postdoctoral Research and Career Symposium, Argonne National Laboratory, poster presentation.
10. **Jung, J.**, Towne, A. (2024). Resolvent-based estimation and control of airfoil wakes. APS Division of Fluid Dynamics, oral presentation.
11. **Jung, J.**, Towne, A. (2024). Toward turbulent wake estimation using a resolvent-based approach. AIAA SciTech Forum.
12. **Jung, J.**, Towne, A. (2023). Resolvent-based estimation for turbulent wakes. APS Division of Fluid Dynamics, oral presentation.
13. **Jung, J.** (2023). Resolvent-based estimation and control of flows, Michigan Fluid Mechanics Research Seminar (internal), oral presentation.
14. **Jung, J.**, Bhagwat, R., Towne, A. (2023). Resolvent-based estimation of laminar flow around an airfoil. AIAA SciTech Forum.
15. **Jung, J.**, Bhagwat, R., Towne, A. (2022). Real-time estimation and control of flow over a NACA0012 airfoil using resolvent-based tools. APS Division of Fluid Dynamics, oral presentation.
16. **Jung, J.** (2021). Resolvent-based flow estimation and control, Michigan Fluid Mechanics Research Seminar (internal), oral presentation.
17. **Jung, J.**, Martini, E., Cavalieri, A., Jordan, P., Lesshafft, L., Towne, A. (2020). Optimal resolvent-based estimation for flow control. APS Division of Fluid Dynamics, oral presentation.

COAUTHORED PRESENTATIONS — PRESENTED BY COLLABORATORS

1. Zhou, Y., Bhagwat, R., **Jung, J.**, Towne, A. (2026). Resolvent-based Estimation and Control of Wavepackets in a Supersonic Jet. APS Division of Fluid Dynamics, oral presentation (*forthcoming*).
2. Zhou, Y., **Jung, J.**, Bhagwat, R., Towne, A. (2025). Estimation and closed-loop control of supersonic jet noise. APS Division of Fluid Dynamics, oral presentation.
3. Constantinescu, E., Kang, S., **Jung, J.** (2025). Hybrid machine learning for multiscale PDE simulation using neural ODEs. 18th U.S. National Congress on Computational Mechanics, oral presentation.
4. Towne, A., **Jung, J.**, Bhagwat, R., Zhou, Y., Paulson, J. (2024). Resolvent-based estimation and control of wakes and jets. 13th International Symposium on Turbulence and Shear Flow Phenomena, oral presentation.
5. Bhagwat, R., **Jung, J.**, Martini, E., Schmidt, O., Jordan, P., Towne, A. (2022). Toward resolvent-based estimation and control of high-speed turbulent jets. APS Division of Fluid Dynamics, oral presentation.
6. Martini, E., **Jung, J.**, Cavalieri, A. V. G., Jordan, P., Towne, A., Lesshafft, L. (2021). Optimal control for large systems: a resolvent-based approach. XXV ICTAM Conference, oral presentation.